

ETHEL SAKYI

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EDUCATION

Winthrop University, Rock Hill- SC

B.S. Candidate Computer Science | Minor in Mathematics and Applied Physics

8/22 – Present

Selected Coursework: Algorithms & Data Structures, Computer Science I & II

Computer Architecture, Operating Systems, Computer Science Theory, Software Engineering I & II, System Administration, Network Processing, Database Processing, Cloud Computing, Web Programming, Mathematical Modelling, Discrete Mathematics, Calculus I & II, Physics with Calculus, Modern Physics, Material Science, Probability & Statistics, Linear Algebra.

RELEVANT EXPERIENCE

Pure Storage

Santa Clara, CA

Technical Program Manager Intern

5/25 – 8/25

- Drove alignment across engineers, QA, and TPMs by defining success criteria and KPIs for release readiness.
- Delivered a release readiness dashboard that gave leadership real-time visibility into product health.
- Authored the full technical design package (system architecture, data flows, API documentation) to guide engineering teams.
- Automated data retrieval with a python script integrating Jira, Jira XRAY, and Jenkins cutting manual tracking by 95% and speeding release decisions.
- Mitigated reliability and scalability risks through detailed cross-solution risk analysis.

Caregiver Mental Wellness Inc

Concord, NC

Student Project Manager and Software Engineer

8/24 – 5/25

- Identified stakeholder pain points by gathering client requirements and translating them into technical specifications.
- Created and tested multiple UI/UX prototypes, iterating with client feedback and engineering input to refine usability.
- Balanced stakeholder needs and technical constraints ensuring that the product design was feasible and aligned with user expectations.
- Designed the database and backend architecture to support secure report storage and generation and provided engineers with implementation documentation.
- Coordinated a 7-person Agile team by setting priorities, delegating tasks, and tracking progress towards milestones
- Delivered a reporting tool that transitioned stakeholders from a manual reporting system to an automated system, saving them 10+ hours a week in report generation,

SC Institute of Biomedical Research Summer Program | Winthrop University

Rock Hill, SC

Undergraduate Student Researcher

5/24 – 6/24

- Developed stochastic simulation programs to model disease containment and assess vaccine efficacy using the firefighter model.
- Explored variations of bootstrap percolation, including the SIR model, to analyze infection and recovery parameters and investigated a probabilistic disease variation.

The Academic Success Center | Winthrop University

Rock Hill, SC

Tutor

5/23 – 1/24

- Tutored students in Microeconomics and Macroeconomics, providing academic support through one-on-one sessions and group discussion.
- Led study sessions and exam preparations in Econometrics to aid students in developing quantitative problem-solving skills and to provide extra support outside of the classroom.

LEADERSHIP EXPERIENCE

DSU Programming Board | Winthrop University

Rock Hill, SC

President

2/24 – Present

- Negotiated with the Board of Trustees amid budget cuts, reinstating ~85% of the organization's original funding to safeguard event programming.
- Restructured committees to reduce redundancy and increase efficiency, ensuring high quality execution even with limited resources.
- Overhauled the event proposal system to streamline approvals and improve accountability.
- Transformed the promotions committee into a data-driven marketing and analytics team, enabling evidence based decisions that improved campaign efficiency and event turnout.
- Established a contract framework and fostered collaborations with clubs and departments, reducing the financial burden whilst expanding programming reach.

Women in Computing Sciences Society | Winthrop University

Rock Hill, SC

Founder and Chair

08/24 – Present

- Established and led the Women in Computer Science club to promote diversity and mentorship within the field.
- Authored the club constitution, recruited members, and built an executive board. Secured a faculty advisor and initiated collaborations with ACM and STARS organizations.
- Developed mentorship programs by engaging female faculty members to support students. Currently working toward official club chartering.

TECHNICAL SKILLS & PROJECTS

Programming: C++, Python, JavaScript, MIPS Assembly Language, SQL, MongoDB, Java, Algorithm Design, Data Structures, Software Testing, Django, React, Figma, Compiler Development, Mathematica, LaTeX, Linux, Unix, Windows, Excel, R, Microsoft Office Suite, Google Sheets, Jira, Confluence, Asana, Grafana, Snowflake, Jenkins, PowerBI.

Management Skills: Agile & SDLC, Cross-Functional Leadership, Stakeholder Engagement, Strategic Roadmapping, Process Optimization, Risk Management, Product Lifecycle Management, Data-Driven Decision Making, Stakeholder Engagement, Risk Analysis, Release Planning, Change Management

Compiler Design Project

Developed a compiler using Lex, Yacc, and Bison to parse and interpret programming languages. Focused on lexical analysis, syntax parsing, and error handling.

Firefighter Simulation

Built a stochastic simulation using Python and NetworkX to model infection spread on a grid graph, analyzing the effectiveness of deploying firefighters to contain spread.

Alexa Skill: Game Facts

Created an Alexa Skill using Node.js to deliver pre-programmed facts about video games. Integrated ASK SDK, tested functionality, and deployed via AWS Lambda.

Bakery Management System

Designed a customer and ticket management system with a MySQL database backend. Implemented secure coding practices, parameterized queries, and CLI functionality for operations like creating, updating, and deleting records.

RESEARCH WORK & EXPERIENCE

CSCI 327 Research Paper | Winthrop University

Rock Hill, SC

Paper Submitted: "Exploring Post-Quantum Cryptography"

2/24

Researched vulnerabilities in classical encryption systems to quantum algorithms, focusing on Shor's Algorithm. Analyzed post-quantum cryptographic methods, including lattice-based, hash-based, and multivariate polynomial cryptography, and examined their implementation challenges.

CSCI 327 Research Paper | Winthrop University

Rock Hill, SC

Paper Submitted: "Mitigating Challenges in Post-Quantum Cryptography Implementation"

4/24

Investigated the computational, integration, and cryptanalysis challenges in deploying post-quantum cryptography. Proposed algorithm optimization strategies, hybrid cryptographic systems, and alternative cryptographic methods using true randomness and biological processes.

Mathematical Association of America Poster Symposium | High Point University

High Point, NC

Research Poster: "Variations of Bootstrap Percolation on the Random Geometric Graph"

4/25

Studied disease spread models using bootstrap percolation on random geometric graphs. Developed and analyzed variations such as the SIR model, Firefighter model, and Coin-Flip Disease Passing model. Presented findings at the MAA-SE conference, highlighting the impact of different parameters on infection dynamics.

ADDITIONAL INFORMATION

Languages: Native bilingual in English & Akan. Working proficiency in French.

Professional Memberships: Re-writing the Code (RTC), National Society of Black Engineers (NSBE), Society of Women Engineers (SWE), American Physical Society (APS), Association of Computing Machinery (ACM).

Others: Palantir Women in Technology Scholarship Recipient, Palantir Student Leaders Conference Attendee.